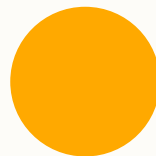




Artificial Intelligence and ML

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AI and ML

- We can write any program now
- And communicate with other programs
- Those are turing complete
- But can we practically write everything?





AI and ML

- Take for instance the following program:
- Respond to a query
- In natural language
- Write a program





AI and ML

- Particularly challenging to write
- Maybe you can read in input?
- But where to go from there?
- Let's think about some other approaches





AI and ML

- This task would be easy for humans to accomplish
- Normally requires human intelligence to solve
- We need to create artificial version





AI and ML

- Artificial Intelligence (AI)
- Not real
- But we can at least give it our best shot
- Create a program that can mimic human intelligence





AI and ML

- Historically, this breaks down into roughly three generations
- We'll talk about each
- But let's start at the beginning
- Alan Turing (hi again!)
- 1949





AI and ML

- As “computing” became a concept
- AI was all the rage
- People wanted to know if AI was even possible
- We didn't have a good grasp of any of this yet
- Needed a way to analyze this





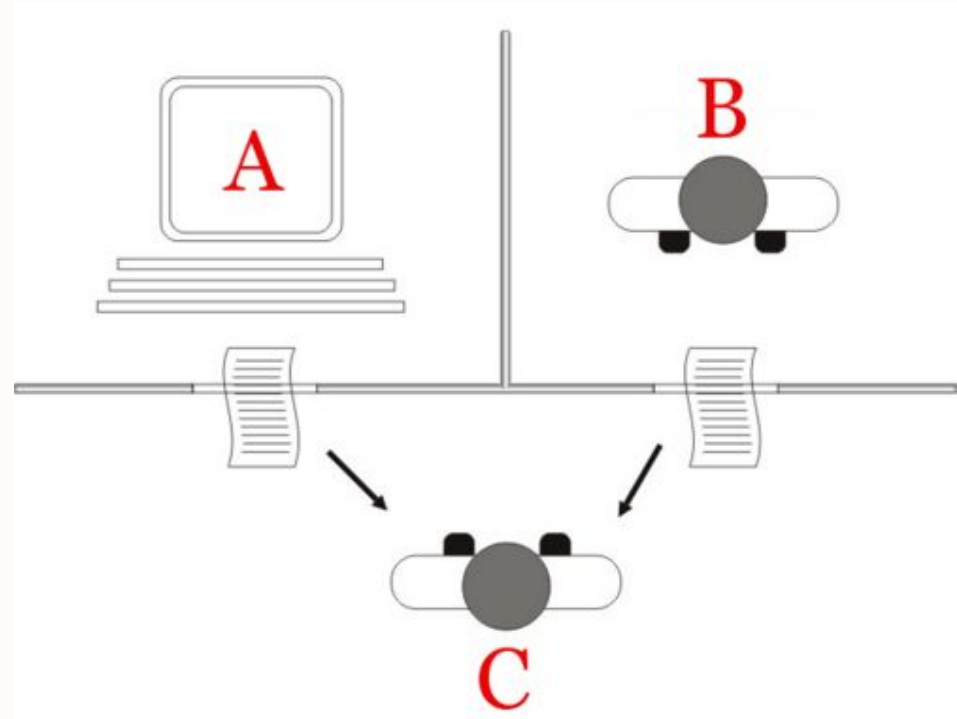
AI and ML

- "Imitation Game"
- Have a person decide if they are talking to a bot
- If the person cannot tell the difference $\Rightarrow$ We have reached AI
- This is an outdated metric



AI and ML

- **Turing Test**
- Ask the machine a series of questions
- Natural Language
- Online Versions





AI and ML

- Early AI
- Rule-Based system
- First Generation
- Remember our if/else statements?
- This is basically early "AI"
- There are many forms of this





AI and ML

- Easiest Approach:
- Just hard-code all of the if-else statements into your program
- Works, but doesn't really scale
- How would this work for natural language?





AI and ML

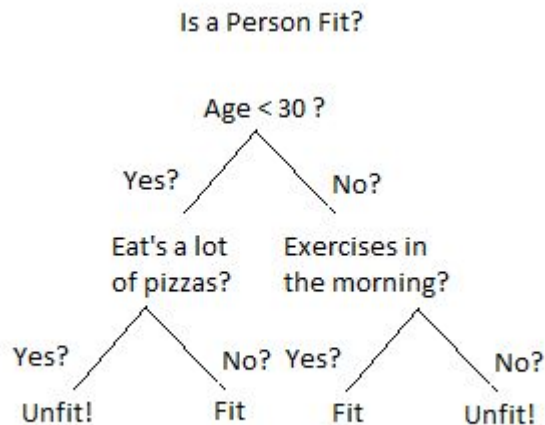
- Different Approach:
- What if we wrote code, that wrote the if statements for us?
- How would that work?
- Write some code
- Loops over some data
- Looks for patterns and easy splits?





AI and ML

- Decision Trees
- *Output* of this process
- Gives you if-else statements





AI and ML

- You would need data to initially find those patterns
- A bunch of math
- Let's see what that data might look like
- Must be labeled
- Data $\Rightarrow$ Output Prediction





AI and ML

- Rigid:
- Doesn't always hold up well to new data
- But, its very easy to understand
- Ease of understanding is an incredibly valuable thing in **ML**





AI and ML

- Depending on your problem domain, these are still used a lot today
- But not perfect
- Lets look at ELIZA
- Not a decision tree, but still rule-based





AI and ML

- Next Generation:
- **Machine Learning (really*)**
- Large Dataset
- Analyze Patterns
- Predict Relationships between data





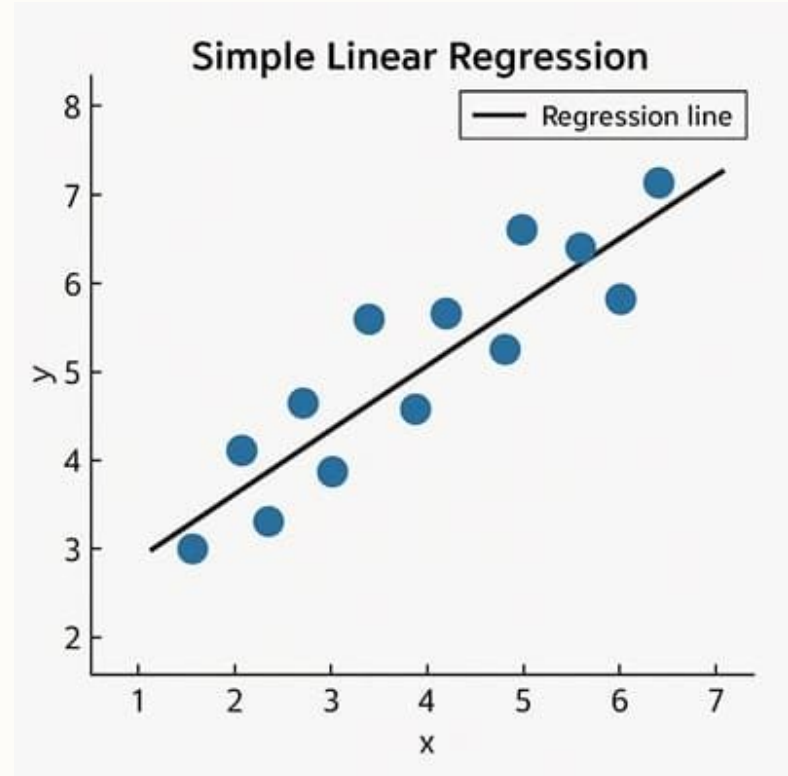
AI and ML

- There is likely some hidden underlying relationship
- Too complex to fully capture
- Predict a simpler relationship between these
- Let's do some math



AI and ML

- **Linear Regression** (from your stats class) is a form of machine learning





AI and ML

- Predicting a simple line from a more complex relationship
- Let's see this

Linear Regression Equation

$$Y = a + bx$$

$$a = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma y)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

$$b = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$





AI and ML

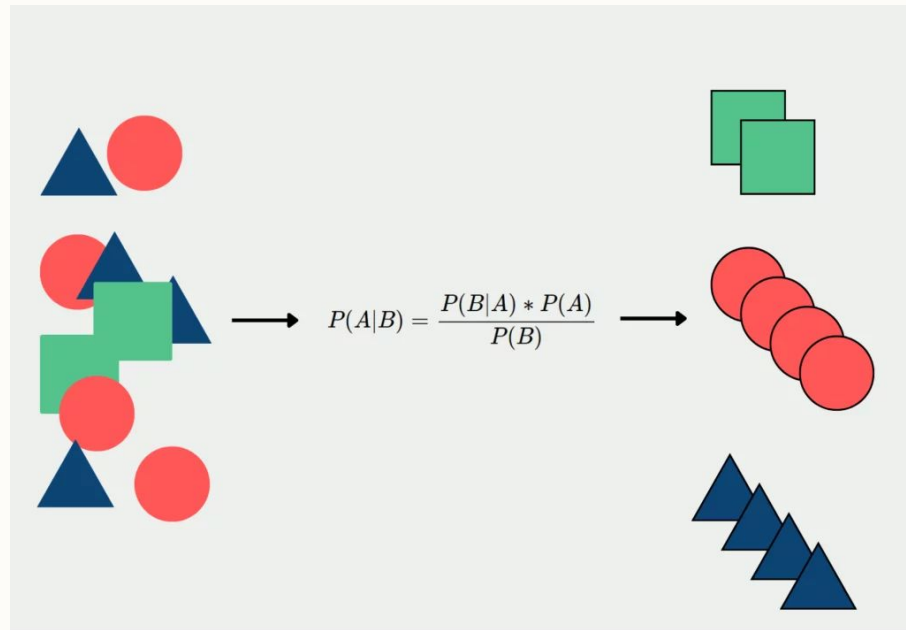
- **Classification**
- Run a similar process, but determine what "class" a point most likely belongs to
- **Naive-Bayes** is a common classifier





AI and ML

- Stats problem (again)
- Based on Bayes Theorem
- Update beliefs as you are exposed to new information





AI and ML

- **We know the features**
- As in, we know what factors are likely to influence the outcome
- E.g. predicted class, or datapoint
- This is not generally the case
- Although it does make it easier





AI and ML

- Next Generation
- **Deep Learning**
- Machine learning, but the features themselves are learned
- We know there is a relationship, but what predicts that relationship is unknown





AI and ML

- Classification
- Dog vs Cat
- How could you code something that tell the difference?
- Well you would need to know what about an image makes it more dog-like than cat-like





AI and ML

- We can tell immediately
- A program cannot
- Write a program to learn what is it about images that makes them "cat" or "dog" like





AI and ML

- Deep learning allows us to capture more complex relationships within data itself
- At the cost of not being able to tell what's happening explicitly
- How does this work in terms of language processing?





AI and ML

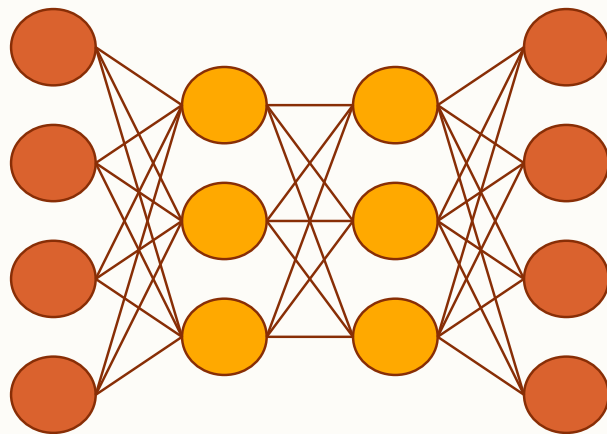
- You are probably more familiar with these
 - "AI" chatbots
 - ChatGPT, Gemini, Claude, etc.
- Attention models
- Transformer-based model





AI and ML

- Based around neural networks
- Encode information as a large, layered graph
- Do the computation as a pass through the graph





AI and ML

- Math is based on the input to the graph
- As well as a series of "weights" assigned to each edge
- Weight is just a number associated with an edge

